Robot-Model-Independent Geometric Descriptor Characterization for Grasp-Oriented Placement Prioritization

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***Abstract*—Quality-diversity (QD) methods such as PLACE IT generate diverse archives of mechanically stable object placements, but stability does not necessarily ensure suitability for subsequent grasping. This paper introduces the Robot-model-independent Geometric Manipulation Proxy (RGMP), a lightweight post-optimization interface for investigating how geometric descriptors influence the prioritization of stable placements for downstream grasp evaluation. RGMP combines reference-frame height, workspace centrality, and uprightness into a Grasp Friendliness Index (GFI), without modifying the original PLACE IT optimizer or archive. Rather than seeking a universally superior ranking policy, this work examines the strengths, limitations, and interactions of individual geometric descriptors and their combination across objects and orientations.**

Franka Panda simulations investigate the contributions of geometric descriptors to downstream grasp success. Centrality-only ranking (tilted poses permitted) achieved 77.8%, 77.8%, 100%, and 85.7% grasp success for Banana, Mug, Power Drill, and Spoon, respectively. Original GFI and centrality-only (upright poses permitted) achieved 66.7%, 40%, 100%, and 85.7% for the same objects. The results show that tilted placements can remain graspable and that upright filtering does not consistently improve grasp success.

To address the limitations of geometric ranking alone, GFI selections were retrospectively screened using recorded Panda static-admissibility information. Among the resulting selected placements, 37/40 (92.5%) achieved grasp success, with object-specific rates of 90%, 90%, 100%, and 90% for Banana, Mug, Power Drill, and Spoon, respectively. This retrospective screening improved selection among previously evaluated placements but does not establish prospective feasibility prediction. Physical TIAGo experiments achieved 93% overall grasp success for GFI-ranked placements versus 42% for randomly selected stable PLACE IT placements.

Additional physical TIAGo Mug experiments compared two near-central configurations: one tilted and one upright, achieving 80% and 70% grasp success, respectively, over ten attempts each. The original GFI-ranked Mug evaluation achieved 90% mean success across ten placements. These individual-pose results do not establish a general advantage of tilted orientations. Gripper-induced object displacement highlights how approach geometry can limit grasp accessibility.

Overall, geometric pre-ranking prioritizes grasp candidates, while robot-specific accessibility remains dependent on orientation and approach geometry.

***Index Terms*—Quality-diversity optimization, robotic manipulation, object placement, grasp planning, geometric pre-ranking.**

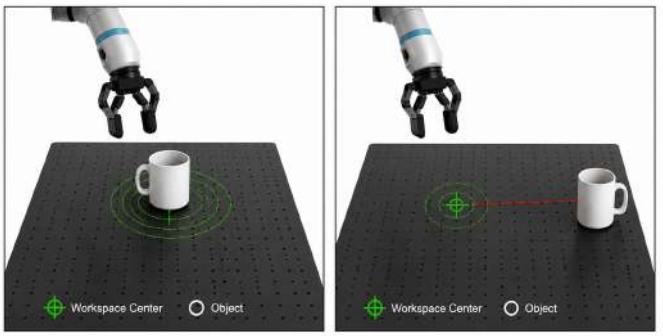


Fig. 1. Mechanically stable placements may differ in grasp accessibility. Geometric pre-ranking prioritizes candidates for downstream grasp evaluation.

I. Introduction

Robotic placement is fundamental to pick-and-place, assembly, sorting, and logistics [1]–[4]. Placement planners seek mechanically stable poses satisfying contact and environmental constraints [5], [6]. However, stability does not guarantee graspability: a stable object may be poorly positioned or oriented for subsequent manipulation (Fig. 1). Grasp synthesis considers geometry, contact quality, collision avoidance, and robot reachability [7]–[10]. Evaluating these constraints across large placement archives can be computationally demanding, motivating lightweight geometric pre-ranking.

Quality-Diversity (QD) methods preserve diverse, high-performing solutions [11], [12]. PLACE IT uses CMA-MAE and physics-based evaluation to archive stable placements [13], [14]. Building on this archive, we introduce the Robot-model-independent Geometric Manipulation Proxy (RGMP), which post-ranks archived 6-DoF placements using a Grasp Friendliness Index (GFI) combining height, centrality, and uprightness. Unlike placement-generation approaches, RGMP does not modify the QD optimizer, stability criteria, or archive; it characterizes how geometric descriptors prioritize placements for subsequent grasp evaluation.

Beyond scoring, this work investigates descriptor strengths and limitations through ablation, robustness tests, Panda simulations, and physical TIAGo experiments. RGMP requires no robot kinematics, gripper geometry, or contact models, but depends on reference-frame conventions. Centrally positioned tilted objects may remain graspable, whereas upright poses can cause gripper interference, motivating retrospective assessment of robot-specific static admissibility.

RGMP is evaluated using six YCB objects [15] through geometric robustness analyses, Franka Panda simulations, and physical TIAGo experiments. Panda evaluations compare GFI with centrality-only (tilted poses permitted), height-only, uprightness-only, height–centrality, and centrality-only (upright poses permitted), followed by retrospective static-admissibility analysis. Physical TIAGo experiments compare GFI against randomly selected stable PLACE IT placements and additionally examine central tilted and upright Mug configurations to investigate orientation-dependent grasp outcomes.

The objective is not to establish that GFI outperforms individual geometric descriptors, but to characterize how descriptor choice and orientation constraints affect placement prioritization. Their usefulness depends on object geometry and gripper approach conditions. We therefore investigate individual descriptors, their combinations, and orientation-dependent physical configurations to identify the strengths and limitations of geometric pre-ranking.

The main contributions are:

1) **Geometric QD extension:** A lightweight RGMP interface for ranking stable PLACE IT placements without modifying CMA-MAE or the original archive.
2) **Geometric characterization:** A systematic investigation of the strengths and limitations of height, centrality, uprightness, and their combinations through component ablation, parameter sensitivity, pose perturbation, and physics-based robustness analysis, emphasizing object-dependent descriptor behavior rather than universal ranking superiority
3) **Panda grasp evaluation:** An empirical comparison of geometric prioritization policies to characterize how descriptor choice and orientation constraints affect grasp outcomes, complemented by retrospective robot-specific static-admissibility analysis.
4) **Physical TIAGo validation:** GFI versus random stable-placement evaluation, complemented by central tilted and upright Mug experiments investigating grasp accessibility and gripper–object interaction.

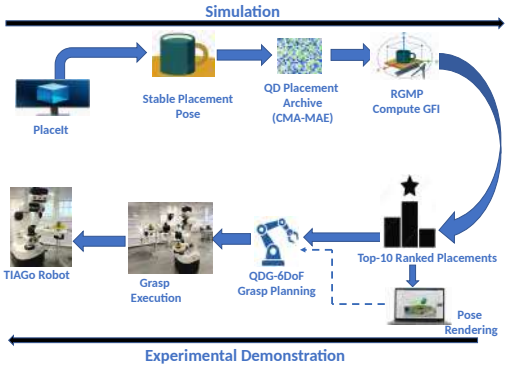


Fig. 2. RGMP extends PLACE IT through geometric GFI ranking of stable archived placements. Top-K candidates undergo downstream QDG-6DoF grasp planning and TIAGo validation. Panda experiments additionally examine retrospective static admissibility.

The remainder of this paper is organized as follows. Section II reviews related work; Section III presents RGMP; Section IV describes the experimental methodology; Section V reports the results; Section VI discusses the findings; Section VII addresses reproducibility and data availability; and Section VIII concludes the paper.

II. Related Work

Robotic placement planning addresses stability, contact equilibrium, and environmental constraints [4], [6], [16], while task-oriented methods consider downstream manipulation [5], [17], [18].

Grasp synthesis assesses contact quality and feasible 6-DoF grasps [7]–[10], [19]. QDG-6DoF applies Quality-Diversity (QD) principles to grasp generation [20].

QD methods preserve diverse, high-performing solutions [11], [12]. PlaceIt uses CMA-MAE [13] and physics to archive stable placements [14], without explicitly optimizing downstream graspability.

Unlike approaches integrating manipulation constraints into planning, RGMP separates geometric prioritization from robot-specific feasibility assessment while preserving PlaceIt's original QD optimization. Beyond weighted scoring, it systematically investigates descriptor strengths, failures, and the role of robot-specific admissibility in refining placement selection.

III. RGMP Framework

RGMP ranks stable PLACE IT placements using GFI without modifying its optimization or archive [13], [14]. Robot-specific grasp planning follows through QDG-6DoF [20] (Fig. 2).

A. Input Representation and Grasp Friendliness Index

RGMP ranks eligible PLACE IT poses [14] using height, centrality, and uprightness (Fig. 3):

$$\mathbf{p} = [x, y, z, \phi, \theta, \psi]^T, \tag{1}$$

where positions are in metres, Euler angles in radians, and the workspace centre is $(0, 0)$. Eligible poses have finite coordinates and $z > 0$.

With $d = \sqrt{x^2 + y^2}$,

$$\begin{aligned} H &= \frac{z}{L_H}, \\ D &= \exp\left(-\frac{d}{L_D}\right), \\ U &= \exp\left(-\frac{\sqrt{\phi^2 + \theta^2}}{\Theta_0}\right), \end{aligned} \tag{2}$$

where $L_H = 1\,\mathrm{m}$, $L_D = 0.1\,\mathrm{m}$, and $\Theta_0 = 1\,\mathrm{rad}$.

$$\boxed{\mathrm{GFI}(\mathbf{p}) = 0.5H + 0.3D + 0.2U.} \tag{3}$$

Weights are fixed heuristics. Height measures reference-point elevation, centrality measures planar distance, and uprightness excludes yaw. GFI is not necessarily bounded by unity and does not encode robot-specific grasp feasibility.

B. Weight Selection and Sensitivity

The GFI weights $(0.5, 0.3, 0.2)$ encode a fixed geometric preference for height, centrality, and uprightness, respectively. Height receives the largest nominal weight to favour elevated object poses, while centrality promotes workspace proximity and uprightness penalizes orientation deviations. These weights are heuristic, not optimized from grasp-success data.

The weights define one fixed geometric preference for the experiments; they are not selected to maximize grasp success and should not be interpreted as establishing an optimal trade-off among descriptors.

Their influence is examined analytically and through sensitivity experiments. Centrality satisfies

$$D(0) = 1, \quad D(L_D) = e^{-1}, \quad D(2L_D) = e^{-2}. \tag{4}$$

Its contribution varies with distance as

$$\frac{\partial \mathrm{GFI}}{\partial d} = -\frac{0.3D}{L_D}. \tag{5}$$

For placements i and j,

$$\begin{aligned} \Delta\mathrm{GFI}_{ij} = 0.5(H_i - H_j) + 0.3(D_i - D_j) \\ + 0.2(U_i - U_j). \end{aligned} \tag{6}$$

GFI is used here as a composite geometric prioritization score, not as a learned or validated predictor of grasp success. Its components express distinct geometric preferences: reference-frame height, workspace centrality, and roll–pitch uprightness. The purpose of combining them is to investigate how these preferences jointly affect placement ranking, not to assume that their weighted sum is universally better than any individual descriptor. Individual descriptor rankings are therefore retained as experimental comparisons.

Thus, a 0.1 centrality gain compensates for a 0.06 normalized-height decrease at fixed uprightness. Actual descriptor influence also depends on archive dispersion:

$$\sigma_{w_X X} = |w_X| \sigma_X. \tag{7}$$

For the Mug archive,

$$\begin{aligned} \sigma_{0.5H} &= 0.0013, \\ \sigma_{0.3D} &= 0.0610, \\ \sigma_{0.2U} &= 0.0683. \end{aligned} \tag{8}$$

Despite its largest coefficient, height has the smallest weighted dispersion. Thus, nominal weights do not directly determine ranking influence.

Importantly, four alternative weight configurations preserved all ten selected placement identities across the six objects, although their ordering varied (Section V). This demonstrates shortlist robustness within the tested weight range, rather than weight optimality or guaranteed grasp performance.

C. Ranking Procedure and Computational Cost

For the eligible archive,

$$\mathcal{A} = \{\mathbf{p}_1, \ldots, \mathbf{p}_N\}. \tag{9}$$

Scores $s_i = \mathrm{GFI}(\mathbf{p}_i)$ are ordered as

$$s_{\pi(1)} \geq s_{\pi(2)} \geq \cdots \geq s_{\pi(N)}, \tag{10}$$

yielding

$$\mathcal{A}_K = \{\mathbf{p}_{\pi(1)}, \ldots, \mathbf{p}_{\pi(K)}\}. \tag{11}$$

Deterministic tie-breaking ensures reproducibility. Computational cost is $O(N)$ for scoring and $O(N \log N)$ for sorting.

D. Retrospective Robot-Specific Static Admissibility

Recorded Panda outcomes define retrospective static admissibility:

$$\boxed{A_R(\mathbf{p}_i) = \begin{cases} 1, & c_i \in \mathcal{S}_{\mathrm{pass}}, \\ 0, & c_i \in \mathcal{S}_{\mathrm{fail}}, \end{cases}} \tag{12}$$

where c_i is the evaluation status and R denotes the robot configuration.

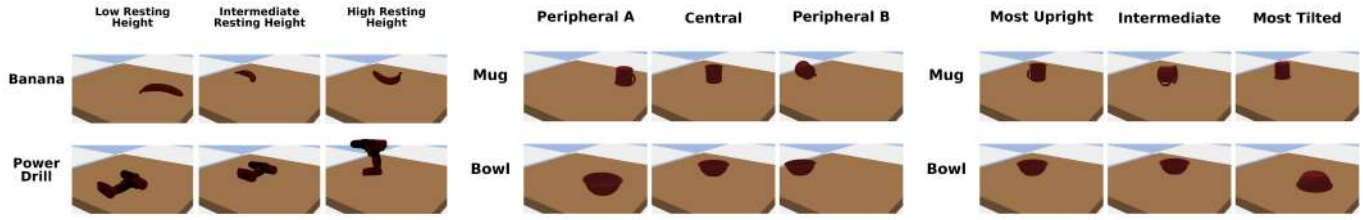
$$\begin{aligned} \mathcal{S}_{\mathrm{pass}} = \{&\texttt{SUCCESS}, \\ &\texttt{APPROACH_FAILED}, \\ &\texttt{GRASP_FAILED}\}. \end{aligned} \tag{13}$$

$$\begin{aligned} \mathcal{S}_{\mathrm{fail}} = \{&\texttt{NO_GEOMETRIC_CANDIDATE}, \\ &\texttt{NO_STATIC_FEASIBLE_CANDIDATE}\}. \end{aligned} \tag{14}$$

Passing indicates static candidate availability, not successful grasp execution.

$$\mathcal{A}_R = \{\mathbf{p}_i \in \mathcal{A}_{\mathrm{tested}} \mid A_R(\mathbf{p}_i) = 1\}. \tag{15}$$

$$\boxed{\mathcal{A}_{R,K}^{\mathrm{GFI}} = \mathrm{TopK}_{\mathbf{p} \in \mathcal{A}_R} \mathrm{GFI}(\mathbf{p}).} \tag{16}$$



(a) Height H (b) Centrality D (c) Uprightness U

Fig. 3. GFI geometric descriptors: reference-frame height, workspace centrality, and roll–pitch uprightness.

TABLE I
SIMULATION PARAMETERS AND PANDA EVALUATION SAMPLES

Parameter	Value
Physics engine / QD algorithm	PyBullet / CMA-MAE
Population / evaluations per object	20 / 2160
Genotype / behaviour descriptor	6-DoF / final pose
QD benchmark objects	6 YCB objects
Panda evaluated placements	154 across 5 objects
Spoon / Banana / Power Drill	24 / 29 / 35
Mug / Bowl	34 / 32
Panda comparative figure	4 objects
Panda physics frequency	240 Hz
Panda holding interval	480 frames
Panda selection budgets	$K = 1, 3, 5, 10$
Uprightness threshold	$U \geq 0.99$

Only candidate eligibility changes; GFI and PLACE IT remain unchanged. Admissibility is reconstructed from prior outcomes, not predicted prospectively.

IV. EXPERIMENTAL METHODOLOGY

A. Placement Archives and Geometric Analyses

PLACE IT generates stable placements using CMA-MAE and PyBullet [13], [14]. Six YCB objects [15] yield 2,091 eligible archived placements. Simulation settings are given in Table I.

Analyses include component ablation, weight sensitivity, pose perturbation, settling, and force disturbances. Sensitivity uses 196 placements, while Panda evaluates 154 unique placements after excluding 31 Plate and 11 development placements. Four-object comparisons exclude Bowl.

B. Geometric Ranking Comparisons

Six geometric policies are compared: GFI, centrality-only (tilted poses permitted), height-only, uprightness-only, height–centrality, and centrality-only (upright poses only), at $K \in \{1, 3, 5, 10\}$.

$$\begin{aligned} S_D &= D, \\ S_H &= H, \\ S_U &= U, \\ S_{HD} &= 0.5H + 0.3D. \end{aligned} \tag{17}$$

Upright-filtered centrality uses

$$\mathcal{A}_U = \{\mathbf{p} \in \mathcal{A} \mid U(\mathbf{p}) \geq 0.99\}. \tag{18}$$

$$\mathcal{A}_K^{DU} = \mathrm{TopK}_{\mathbf{p} \in \mathcal{A}_U} D(\mathbf{p}). \tag{19}$$

The GFI height ablation is

$$\mathrm{GFI}_{DU} = 0.3D + 0.2U. \tag{20}$$

Sensitivity tests vary centrality scale

$$L_D \in \{0.05, 0.075, 0.10, 0.125, 0.15, 0.20\}\,\mathrm{m},$$

and GFI weights:

$$\begin{aligned} &(0.45, 0.35, 0.20), \quad (0.55, 0.25, 0.20), \\ &(0.45, 0.30, 0.25), \quad (0.55, 0.30, 0.15), \\ &(0.40, 0.35, 0.25), \quad (0.60, 0.25, 0.15). \end{aligned}$$

These variations assess ranking sensitivity, not coefficient optimality.

C. Pose Perturbation and Disturbance Tests

Pose robustness is evaluated at perturbation levels $(2\,\mathrm{mm}, 1^\circ)$, $(5\,\mathrm{mm}, 3^\circ)$, and $(10\,\mathrm{mm}, 5^\circ)$, with 100 repetitions per object. Top-10 retention measures ranking sensitivity to pose variations.

Across 1,440 trials, forces along $\pm X$ and $\pm Y$ at 10–40% of object weight assess rotations exceeding 1°.

D. Independent Panda Grasp Simulation

Panda evaluates 154 unique placements: Spoon (24), Banana (29), Power Drill (35), Mug (34), and Bowl (32), selected from five original ranking policies: GFI, centrality-only (tilted poses permitted), height-only, uprightness-only, and height–centrality. A sixth geometric policy, upright-filtered centrality, was evaluated retrospectively without additional simulations. Duplicate selections share recorded outcomes. The evaluation subset is policy-derived, not randomly sampled from the full archives.

Grasps undergo kinematic, collision, approach, closure, lift, and retention checks. Candidate limits are 40 (Bowl: 20); gripper openings are 30 mm for Spoon/Banana and 80 mm otherwise. Simulation runs at 240 Hz with 480 holding frames.

Success requires approach error below 3 mm, finger contact, lift above 25 mm, and slip below 10 mm and 15°.

Four-object retrospective comparisons assess seven conditions: six geometric policies and GFI with static-admissibility screening (Section III-D). All comparisons reuse recorded Panda outcomes.

For contact positions $\mathbf{A}, \mathbf{B}$ and normals $\mathbf{n}_A, \mathbf{n}_B$,

$$\mathbf{d}_c = \frac{\mathbf{B}-\mathbf{A}}{\|\mathbf{B}-\mathbf{A}\|},$$
$$a = -\mathbf{n}_A^T \mathbf{d}_c,$$
$$b = \mathbf{n}_B^T \mathbf{d}_c.$$

For $a, b > 0$, the antipodal margin is

$$m_{\text{anti}} = 0.5 - \max\left\{\frac{\sqrt{1-a^2}}{a}, \frac{\sqrt{1-b^2}}{b}\right\}. \tag{21}$$

With friction coefficient 0.5, it measures contact alignment rather than complete grasp quality [19].

E. Physical TIAGo Experiment

TIAGo experiments evaluate GFI ranking and Mug orientation using three YCB objects [15].

1) GFI Versus Random Stable Placements: For each object, ten highest-GFI placements are compared with ten stratified-random stable placements. Random selection uses five planar-distance quantile bands, sampling two placements per band without replacement, independently of GFI. Ten attempts per manually reconstructed placement yield 600 trials (Fig. 4).

Placement-level grasp success is

$$\text{GSR}_r = \frac{N_r^{\text{success}}}{N_r^{\text{attempts}}} \times 100\%. \tag{22}$$

Archived poses are manually reconstructed on a marked workspace; reconstruction errors were not measured, limiting pose-level reproducibility. An object-stratified placement bootstrap (20,000 repetitions; seed 2027) estimates uncertainty. Within-object Spearman correlations assess GFI–success associations. Post-hoc centrality comparisons do not constitute new physical trials.

2) Additional Physical Mug Orientation Validation: Two near-central Mug poses are tested: tilted (index 125; $85.7°$),

$$(x, y)_{\text{tilted}} = (0.000432, -0.000559)\,\text{m}. \tag{23}$$

and upright (index 901; approximately $0°$),

$$(x, y)_{\text{upright}} = (-0.000003, -0.000015)\,\text{m}. \tag{24}$$

Coordinates use the PLACE IT frame. Each manually reconstructed pose receives ten TIAGo grasp attempts, adding 20 trials.

Results are compared descriptively with the original GFI Mug evaluation. The upright pose also belongs to the GFI top-ten shortlist. Gripper-induced displacement is noted where observed.

V. Results

A. Archive Composition and Component Ablation

The six PLACE IT archives contain 2,091 stable placements: Mug (481), Bowl (217), Spoon (389), Power Drill (549), Plate (242), and Banana (213) (Fig. 5).

Table II shows that centrality and uprightness correlate more strongly with GFI than height, but shortlist overlap differs.

TABLE II
GFI Component Ablation Across Six Objects

Component	Top-10 overlap	Mean ρ
Height (H)	0.0/10	-0.499
Centrality (D)	6.5/10	0.600
Uprightness (U)	0.5/10	0.630

Height-only has zero top-ten overlap with GFI; this does not imply lower grasp success.

B. Sensitivity and Robustness

All tested weight variants preserved top-ten membership across six objects, although ordering changed.

Figure 6 compares orientation and settling across ranking methods.

Mean top-ten retention under $(2\,\text{mm}, 1°)$, $(5\,\text{mm}, 3°)$, and $(10\,\text{mm}, 5°)$ perturbations was 9.03, 7.43, and 5.62, respectively (100 repetitions per object).

Across 1,440 disturbance trials ($\pm X$, $\pm Y$; 10–40% of object weight), no GFI-ranked placements exceeded $1°$ rotation versus 11/60 random placements (difference: 18.33 pp; 95% CI: $[11.67, 26.67]$; Fig. 8). This evaluates placement stability and mechanical robustness, not robot-specific graspability.

C. Panda Candidate Admissibility and Grasp Outcomes

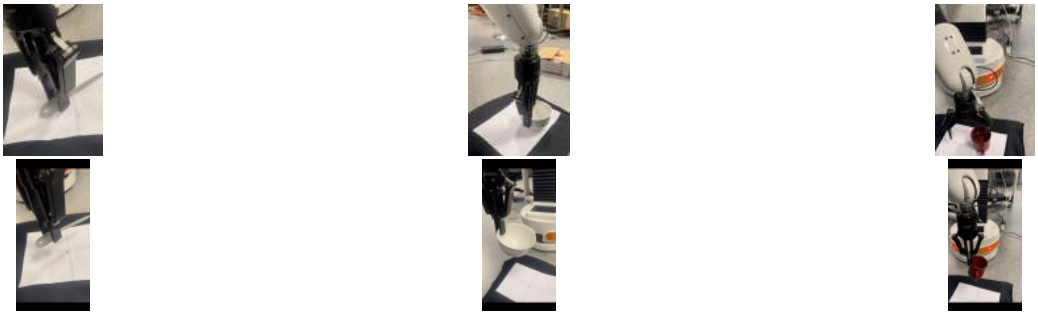
Of 154 Panda placements, 130 had static grasp candidates and 101 succeeded; 24 lacked static candidates, six failed approach, and 23 failed grasp/retention. The four-object comparisons exclude Bowl.

a) Matched-budget comparison.: At $K = 10$ (Fig. 9), the four-object comparison uses ten selection slots per policy and object. Original GFI achieved 30/40 (75%), centrality-only (tilted poses permitted) 35/40 (87.5%), height-only 29/40 (72.5%), uprightness-only and height–centrality 35/40 (87.5%) each, and centrality-only (upright poses permitted) 30/40 (75%).

Retrospective static-admissibility screening increased GFI successful selections from 30/40 (75%) to 37/40 (92.5%) among previously evaluated Panda placements. This improvement reflects recorded robot-specific admissibility information, not prospective feasibility prediction or GFI alone. Successful approach, grasp execution, and retention are not guaranteed.

This result illustrates how robot-specific candidate availability can refine geometric prioritization. It is not evidence that GFI itself predicts grasp feasibility or that its geometric descriptors outperform centrality-only ranking.

These policies are compared to characterize descriptor-dependent placement prioritization, not to identify a universally optimal ranking policy. The results illustrate how geometric descriptor choice affects placement selection, while robot-specific static admissibility can refine retrospective selection. They do not establish that GFI is superior to centrality-only ranking or that geometric descriptors alone predict grasp feasibility.



(a) Spoon (b) Bowl (c) Mug

Fig. 4. Physical TIAGo validation of GFI-ranked placements: manually reconstructed stable placements (top) and representative successful robotic lifts (bottom) for Spoon, Bowl, and Mug. The original experiment compares GFI-ranked placements against randomly selected stable PLACE IT placements.



(a) GFI distribution (b) PLACE IT fitness versus GFI

Fig. 5. Archive-level comparison across six YCB objects: (a) GFI distributions and (b) PLACE IT fitness versus GFI.

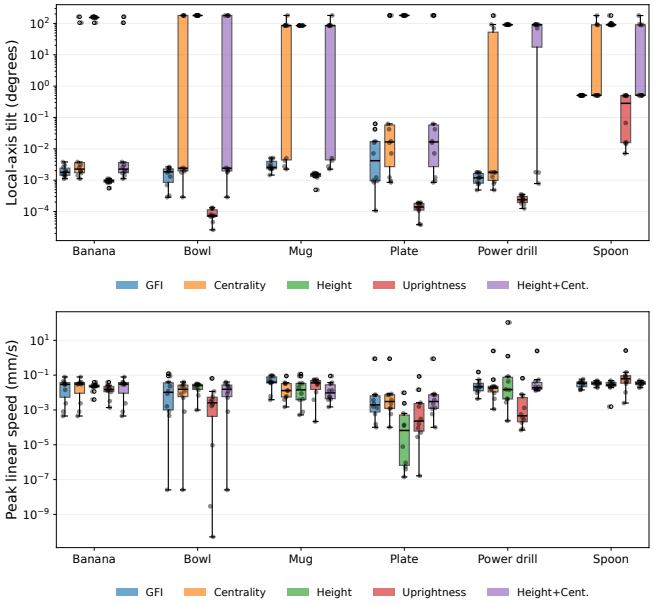


Fig. 6. Geometric characteristics of five ranking methods across six objects: (a) local-axis tilt and (b) peak residual linear speed, shown on logarithmic scales.

b) Object-specific shortlist outcomes.: Unlike the matched-budget comparison, Fig. 10 reports object-specific evaluated subsets with varying denominators. Centrality-only (tilted poses permitted) versus original GFI achieved 7/9 (77.8%) versus 6/9 (66.7%) for Banana, 7/9 (77.8%) versus 4/10 (40%) for Mug, 10/10 (100%) each for Power Drill, and 6/7 (85.7%) each for Spoon.

Centrality-only (upright poses permitted) matched original

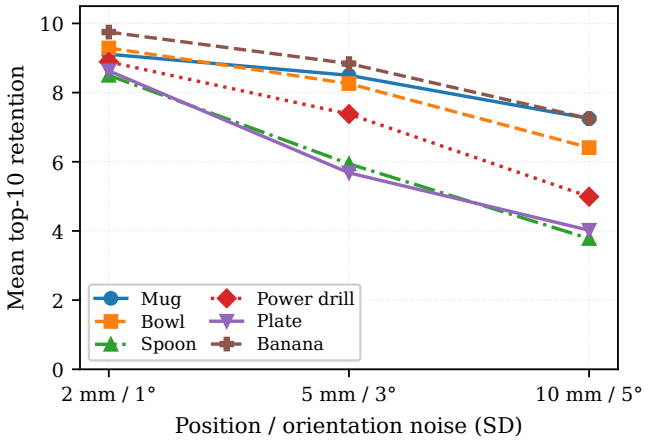


Fig. 7. Robustness of top-10 placement selection to translational and angular pose perturbations across six YCB objects. Each curve represents the mean number of retained placements over 100 repetitions per object at each noise level.

GFI success rates. Retrospectively screened GFI achieved 9/10 (90%), 9/10 (90%), 10/10 (100%), and 9/10 (90%), respectively. All comparisons reuse recorded Panda outcomes.

Unlike Fig. 9, which re-ranks evaluated candidates, Fig. 10 uses archive-based top-ten shortlists. Unevaluated placements are excluded from the denominators, not counted as failures. Overlapping selections reuse recorded outcomes.

Centrality-only can grasp tilted placements; uprightness does not guarantee success. Static admissibility affects retrospective selection, not subsequent approach or execution feasibility.

c) Interpretation of descriptor comparisons.: The ranking comparisons are intended to characterize the effects of

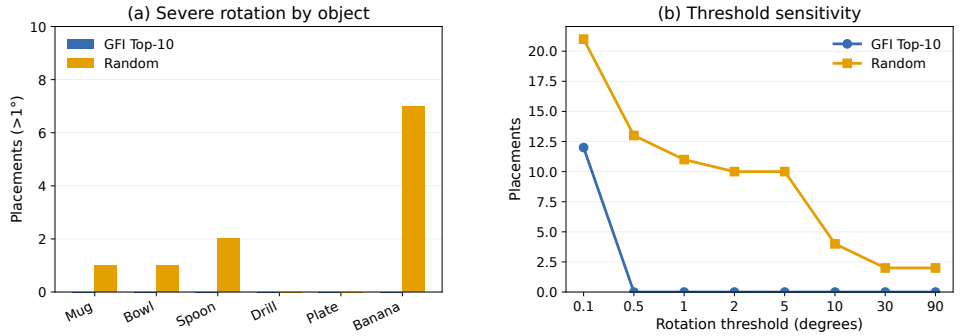


Fig. 8. Simulated disturbance robustness across six objects: (a) placements with maximum rotation greater than 1°; (b) placements exceeding different rotation thresholds. Counts represent the maximum rotation across 12 disturbance conditions for 60 GFI-ranked and 60 random placements.

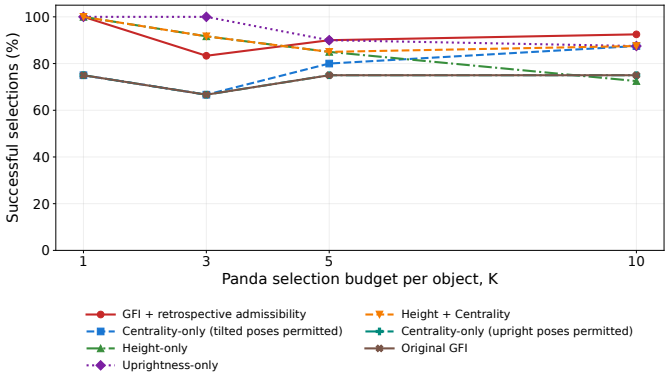


Fig. 9. Retrospective Franka Panda successful-selection fractions across Banana, Mug, Power Drill, and Spoon at matched budgets $K \in \{1, 3, 5, 10\}$. Seven conditions are shown: original GFI, centrality-only (tilted poses permitted), height-only, uprightness-only, height–centrality, centrality-only (upright poses permitted; $U \geq 0.99$), and GFI with retrospective static-admissibility screening. Only the last curve conditions selection on recorded robot-specific static-candidate availability; no additional Panda simulations were performed.

geometric descriptor choice, rather than to establish a universally superior policy. Centrality-only ranking permits tilted placements and, for some objects, prioritizes configurations that remain graspable despite their orientation. Restricting centrality to upright configurations changes the candidate set and can reduce grasp success, illustrating that uprightness is not a universally beneficial constraint. GFI expresses a different geometric preference by combining height, centrality, and uprightness. The observed differences therefore motivate examining descriptor interactions and object-dependent grasp accessibility; they do not establish that GFI is superior to centrality-only or that tilted placements are generally preferable.

D. Physical TIAGo Results

a) GFI versus random stable placement.: Across 30 placements per policy (ten attempts each), TIAGo achieved (93%) GFI successes versus (42%) for random stable placements: +51.0 pp (95% placement-bootstrap CI $[35.0, 66.7]$ pp).

$$\begin{aligned}\Delta_{\mathrm{GSR}} &= \overline{\mathrm{GSR}}_{\mathrm{RGMP}} - \overline{\mathrm{GSR}}_{\mathrm{Rand}} \\ &= 93.0\% - 42.0\% = 51.0 \text{ pp}.\end{aligned} \tag{25}$$

TABLE III
PHYSICAL TIAGO GFI VERSUS RANDOM STABLE PLACEMENT GRASP SUCCESS

Object	GFI (%)	Random (%)	Δ (pp)
Mug	90	36	54
Bowl	96	47	49
Spoon	93	43	50

Overall difference: 51.0 pp; placement-bootstrap 95% CI: $[35.0, 66.7]$ pp.

GFI/random success for Mug, Bowl, and Spoon was 90/36%, 96/47%, and 93/43%, respectively (Table III; Fig. 12).

$$\Delta^{(o)}_{\mathrm{GSR}} = \overline{\mathrm{GSR}}^{(o)}_{\mathrm{GFI}} - \overline{\mathrm{GSR}}^{(o)}_{\mathrm{Rand}}. \tag{26}$$

The CI used 20,000 object-stratified placement bootstrap resamples (seed 2027). Given the small physical sample, it does not establish generalization to unseen placements or objects or account for reconstruction and execution variability. Within the GFI top-ten, success correlations were 0.9342 (Mug), 0.8528 (Bowl), and 0.9211 (Spoon); these do not generalize to full archives.

b) Additional central Mug validation.: The additional Mug experiments were designed to investigate whether orientation-dependent grasp accessibility could help explain the Panda descriptor results. They compare two near-central configurations, one tilted and one upright, rather than evaluating complete centrality-only and upright-filtered centrality ranking policies. The purpose is to examine grasp execution and gripper–object interaction at selected configurations, not to establish a general ranking-policy advantage. Additional Mug trials found 8/10 (80%) grasps for a central tilted pose (index 125; 85.7°) and 7/10 (70%) for a central upright pose (index 901; 0°). The original GFI Mug top-ten achieved 90/100 (90%) (Fig. 13).

These single-pose trials do not establish orientation superiority. Gripper-induced Mug displacement illustrates approach-clearance limitations; the upright pose also belongs to the GFI shortlist.

Overall, RGMP improves upon the tested random baseline, while grasp success remains robot- and contact-dependent.

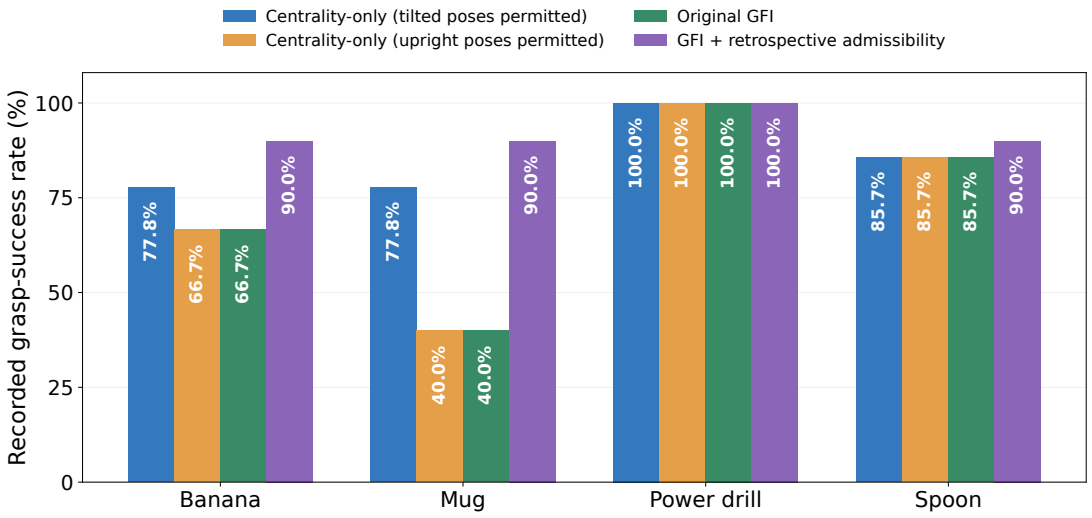


Fig. 10. Franka Panda grasp-success rates for centrality-only (tilted poses permitted), centrality-only (upright poses permitted), original GFI, and GFI with retrospective static admissibility. Rates use recorded outcomes; unevaluated selections are excluded.

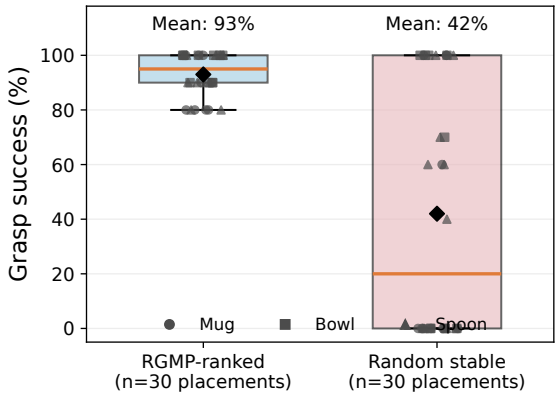


Fig. 11. Placement-level physical TIAGo grasp-success distributions for GFI-ranked and randomly selected stable placements ($n = 30$ per group). Mean difference: 51.0 pp (95% placement-bootstrap CI: $[35.0, 66.7]$ pp).

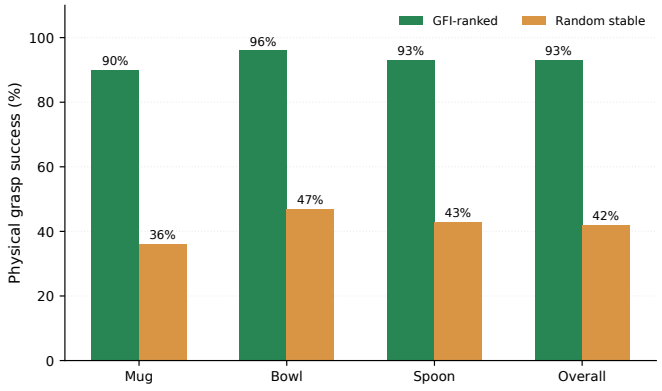


Fig. 12. Physical TIAGo grasp-success rates for GFI-ranked and randomly selected stable PLACE IT placements across Mug, Bowl, and Spoon. Overall rates were 93% (279/300) for GFI and 42% (126/300) for random stable placements.

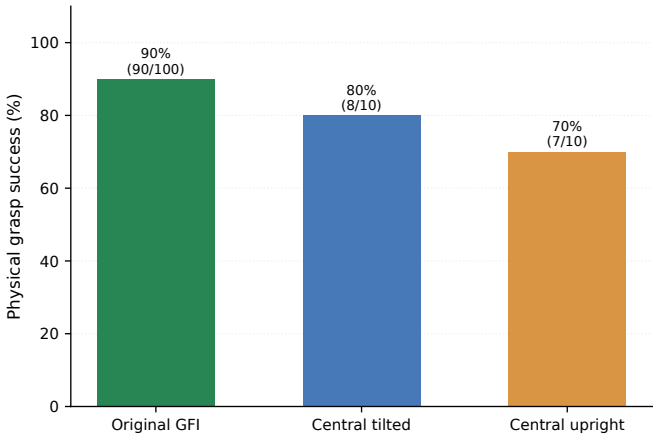


Fig. 13. Physical TIAGo Mug grasp-success comparison: original GFI top-ten placements (90/100, 90%), a single central tilted configuration (8/10, 80%), and a single central upright configuration (7/10, 70%). The latter two conditions involve ten attempts on one pose each and investigate orientation-dependent grasp accessibility and gripper–object interaction. They do not constitute a complete centrality-only top-ten evaluation or a ranking-policy comparison.

VI. Discussion

The Panda results demonstrate that geometric descriptors alone do not guarantee grasp feasibility. At $K = 10$, centrality-only ranking, which permits tilted placements, upright-filtered centrality, and original GFI achieved grasp-success rates of 87.5%, 75%, and 75%, respectively. These findings indicate that restricting central placements to upright orientations does not necessarily improve grasp success.

Taken together, the descriptor comparisons show that geometric placement preferences are object- and configuration-dependent. Centrality-only ranking can prioritize graspable tilted configurations, while an uprightness constraint can exclude configurations that remain accessible to the gripper. GFI combines these preferences, but the experiments do

not establish that this combination is universally superior to individual descriptors. Instead, the results illustrate why geometric ranking should be understood as a prioritization mechanism whose behavior depends on the descriptors and object geometry. The additional TIAGo Mug trials provide a physical investigation of this orientation-dependent accessibility, while the observed gripper-induced displacement further demonstrates that geometric descriptors do not capture all aspects of grasp execution.

This behaviour highlights the interaction between object orientation, reference-point height, and gripper approach geometry. Although upright placements may be geometrically favourable, they can obstruct the gripper approach or cause unintended object displacement. Consequently, height and uprightness alone cannot represent robot-specific grasp feasibility.

GFI with retrospective static-admissibility screening increased successful selections from 30/40 (75%) to 37/40 (92.5%) among previously evaluated Panda placements. This result reflects recorded grasp outcomes after retrospective screening using robot-specific static-admissibility information. Although the screening demonstrates how recorded candidate availability can refine geometric placement selection, it does not establish prospective grasp-feasibility prediction or guarantee successful approach, grasp execution, or retention.

Physical TIAGo experiments further supported geometric pre-ranking, with GFI-ranked placements achieving 93% grasp success compared with 42% for random stable placements. Additional Mug experiments recorded 80% success for a central tilted configuration and 70% for a central upright configuration, each over ten attempts. These individual-pose results do not establish a general advantage of tilted orientations. Observed gripper-induced Mug displacement further illustrates the limitations of geometric descriptors in representing approach clearance.

These physical results support GFI-based prioritization over the tested random baseline, but do not establish superiority over other ranking policies or generalization beyond the evaluated objects and robot platform.

Overall, GFI outperformed the tested random baseline, but physical validation does not establish superiority over alternative geometric ranking policies. Grasp success remains robot- and contact-dependent. The findings are limited by the tested objects and robot, retrospective outcome coverage, manual reconstruction uncertainty, and the absence of explicit gripper geometry and collision constraints. Future work will investigate prospective admissibility screening and broader physical validation.

VII. Reproducibility and Data Availability

Supporting code, experimental configurations, Panda placement-level records, and analysis scripts are publicly available at https://github.com/aladetola1/Aladetola (commit `af02a862`; directory: `ICARA2027_REPRODUCIBILITY_V3`).

VIII. Conclusion

This paper introduced RGMP, a lightweight, robot-independent geometric pre-ranking extension of PLACE IT that preserves its original optimization framework.

Panda simulations characterized how geometric descriptor choice affects placement prioritization and downstream grasp outcomes. Centrality-only ranking, which permits tilted placements, identified graspable configurations for several objects, while restricting centrality to upright configurations changed the selected placements and could reduce grasp success. These findings demonstrate object-dependent effects of orientation constraints and highlight that descriptor combinations are not universally superior to individual geometric preferences. GFI with retrospective static-admissibility screening increased successful selections from 30/40 (75%) to 37/40 (92.5%) among previously evaluated Panda placements. This improvement does not establish prospective grasp-feasibility prediction.

Physical TIAGo experiments demonstrated the practical value of GFI over random stable placements (93% versus 42%). Additional Mug experiments illustrated how orientation and gripper–object interaction influence grasp accessibility, without establishing a general advantage for tilted poses.

Overall, RGMP complements PLACE IT and robot-specific grasp planning through geometric pre-ranking of stable placements.

The contribution of RGMP is therefore not the demonstration of a universally best geometric ranking, but the systematic characterization of geometric descriptor behavior and the identification of where robot-specific feasibility assessment remains necessary.

Appendix A
Original GFI Rankings and Experimental Traceability

Table IV lists the ten highest original GFI scores for each object; Mug, Bowl, and Spoon were used for TIAGo validation.

TABLE IV
Original top-10 RGMP-ranked GFI values across six YCB objects.

Rank	Mug	Bowl	Spoon	Plate	Banana	Drill
1	.6296	.6235	.6120	.6172	.6193	.6241
2	.6296	.6234	.6120	.6172	.6193	.6241
3	.6296	.6234	.6120	.6172	.6193	.6233
4	.6295	.6171	.6077	.6140	.6193	.6191
5	.6134	.6138	.6021	.6139	.6047	.6046
6	.6079	.6061	.5992	.6136	.5955	.6036
7	.6046	.5817	.5987	.6129	.5847	.6036
8	.5975	.5791	.5958	.6126	.5817	.5992
9	.5961	.5771	.5948	.6085	.5797	.5909
10	.5958	.5656	.5942	.6075	.5779	.5893

Appendix B
Exported Rank-1 Hardware Reference Poses

Tables V and VI report the exported Rank-1 reference poses used for TIAGo validation.

TABLE V
EXPORTED RANK-1 RGMP POSITIONS REPRODUCED DURING HARDWARE VALIDATION.

Object	Rank	GFI	x (m)	y (m)	z (m)
Mug	1	.6296	-.00003	-.00015	.259377
Bowl	1	.6235	-.00005	.00013	.247122
Spoon	1	.6120	-.00039	.00059	.228026

TABLE VI
EXPORTED RANK-1 RGMP ORIENTATIONS REPRODUCED DURING HARDWARE VALIDATION.

Object	Roll (°)	Pitch (°)	Yaw (°)
Mug	-.003	-.001	75.219
Bowl	-.002	.001	121.901
Spoon	.072	.502	144.187

APPENDIX C
RANDOM BASELINE RECORDS

Ten stable placements per object were randomly selected independently of GFI. GFI scores were computed only after selection and did not affect the sampled placements.

Table VII reports the first exported pose for each object. Table VIII lists the post-selection GFI scores for all ten placements.

TABLE VII
FIRST EXPORTED RANDOM-BASELINE POSE PER PHYSICALLY TESTED OBJECT. POSITIONS ARE IN METRES AND ORIENTATIONS IN DEGREES.

Object	GFI	x	y	z	Roll	Pitch	Yaw
Mug	.4693	-.0025	-.0025	.2698	85.60	-.01	128.47
Bowl	.5656	-.0028	-.0213	.2471	-.00	.00	-155.58
Spoon	.5475	-.0001	-.0243	.2280	.07	.50	2.79

TABLE VIII
POST-SELECTION GFI VALUES OF RANDOMLY SELECTED STABLE PLACEMENTS.

Sample	Mug	Bowl	Spoon
1	.4693	.5656	.5475
2	.4023	.5351	.5619
3	.3594	.3338	.5241
4	.3678	.5037	.3254
5	.3549	.4474	.4419
6	.5002	.2583	.4515
7	.2844	.4184	.4214
8	.2605	.4242	.4219
9	.4364	.3991	.3881
10	.2756	.3983	.1707

REFERENCES

[1] O. D. Aladetola, N. O. Adelakun, and A. O. Ajayi, "An innovative pick and place robotic arm system using an ATmega328 microcontroller with escalating stability," *Indian Journal of Advanced Science, Engineering and Technology*, vol. 1, no. 2, pp. 39–52, 2022.

[2] O. D. Aladetola, K. F. Akingbade, and N. O. Adelakun, "Development of a two link robotic manipulator," *Iconic Research and Engineering Journals*, vol. 3, no. 8, pp. 13–21, 2020.

[3] B. Siciliano and O. Khatib, Eds., *Springer Handbook of Robotics*, 2nd ed. Cham, Switzerland: Springer, 2016.

[4] J. A. Haustein, K. Hang, J. A. Stork, and D. Kragic, "Object placement planning and optimization for robot manipulators," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems*, Macau, China, 2019, pp. 7417–7424.

[5] K. Harada, T. Tsuji, K. Nagata, N. Yamanobe, and H. Onda, "Validating an object placement planner for robotic pick-and-place tasks," *Robotics and Autonomous Systems*, vol. 62, no. 10, pp. 1463–1473, 2014.

[6] P. Nadeau and J. Kelly, "Stable object placement planning from contact point robustness," *IEEE Transactions on Robotics*, vol. 41, pp. 3669–3683, 2025.

[7] A. T. Miller and P. K. Allen, "GraspIt! a versatile simulator for robotic grasping," *IEEE Robotics & Automation Magazine*, vol. 11, no. 4, pp. 110–122, Dec. 2004.

[8] J. Mahler, J. Liang, S. Niyaz, M. Laskey, R. Doan, X. Liu, J. A. Ojea, and K. Goldberg, "Dex-Net 2.0: Deep learning to plan robust grasps with synthetic point clouds and analytic grasp metrics," in *Proceedings of Robotics: Science and Systems*, Cambridge, MA, USA, 2017.

[9] H.-S. Fang, C. Wang, M. Gou, and C. Lu, "GraspNet-1Billion: A large-scale benchmark for general object grasping," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 11 444–11 453.

[10] M. Sundermeyer, A. Mousavian, R. Triebel, and D. Fox, "Contact-GraspNet: Efficient 6-DoF grasp generation in cluttered scenes," in *Proceedings of the IEEE International Conference on Robotics and Automation*, 2021, pp. 13 438–13 444.

[11] J.-B. Mouret and J. Clune, "Illuminating search spaces by mapping elites," *arXiv preprint arXiv:1504.04909*, 2015.

[12] J. K. Pugh, L. B. Soros, and K. O. Stanley, "Quality diversity: A new frontier for evolutionary computation," *Frontiers in Robotics and AI*, vol. 3, p. 40, 2016.

[13] M. C. Fontaine and S. Nikolaidis, "Covariance matrix adaptation MAP-annealing," in *Proceedings of the Genetic and Evolutionary Computation Conference*, 2023, pp. 456–465.

[14] A. Ferrad, J. Huber, F. Hélénon, J. Gleyze, M. Khoramshahi, and S. Doncieux, "PlaceIt! a framework for learning robot object placement skills," *arXiv preprint arXiv:2510.09267*, 2025.

[15] B. Calli, A. Singh, A. Walsman, S. Srinivasa, P. Abbeel, and A. M. Dollar, "The YCB object and model set: Towards common benchmarks for manipulation research," in *Proceedings of the International Conference on Advanced Robotics*, Istanbul, Türkiye, 2015, pp. 510–517.

[16] M. T. Mason, *Mechanics of Robotic Manipulation*. Cambridge, MA, USA: MIT Press, 2001.

[17] P. Tournassoud, T. Lozano-Pérez, and E. Mazer, "Regrasping," in *Proceedings of the IEEE International Conference on Robotics and Automation*, vol. 4, 1987, pp. 1924–1928.

[18] K. Fang, Y. Zhu, A. Garg, A. Kurenkov, V. Mehta, L. Fei-Fei, and S. Savarese, "Learning task-oriented grasping for tool manipulation from simulated self-supervision," *The International Journal of Robotics Research*, vol. 39, no. 2–3, pp. 202–216, 2020.

[19] C. Ferrari and J. Canny, "Planning optimal grasps," in *Proceedings of the IEEE International Conference on Robotics and Automation*, Nice, France, 1992, pp. 2290–2295.

[20] J. Huber, F. Hélénon, M. Kappel, E. Chelly, M. Khoramshahi, F. Ben Amar, and S. Doncieux, "Speeding up 6-DoF grasp sampling with quality-diversity," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2024.